

CA/Browser Forum Tracker

State of Affairs Since the September Edition

Last updated: 2026-10-04

About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

This report covers the CA/Browser Forum and the browser root programs from the last full edition, at the start of September, to 4 October 2026. It is told as one story, because one question ran under most of what happened. When the web moves to post-quantum signatures, who will issue its certificates?

The report opens with the few developments that change the picture. It then follows the new kind of certificate authority that Chrome and Apple are designing, and Cloudflare's arrival as a would-be issuer of it. From there it returns to the Forum's own post-quantum ballots, the rules that took effect in September, and the ballots still being argued.

Next comes who the root programs are admitting, and how incumbent authorities fared under new deadlines. The report closes with the picture as it stood before September.

1. What Matters Most

A month of Forum business holds many small changes and a few that alter the shape of the field, and this section names the few.

Four things changed the picture this month, and the first is a new entrant. On 29 September Cloudflare said it intends to become a public certificate authority [1]. It is acquiring publicly trusted root key material from GlobalSign, a deal it expects to close within two months [1]. It has applied to the Chrome, Apple, Microsoft and Mozilla root programs [1]. The company says it will issue both conventional certificates and Merkle Tree Certificates, with standard Merkle issuance offered at no cost [1].

Cloudflare's target is the key detail. It aims for inclusion in Chrome's quantum-resistant root store by early 2027 [1]. That store does not exist in production yet, and Chrome plans its production phase for 2027 [2]. A content delivery network is applying to be trusted under rules that are still in draft [3].

The second change is that Apple joined Chrome on the same design. On 21 September Apple's root program announced a Merkle Tree Certificate policy for web server authentication, with a

draft due by the end of October [4]. Two of the four major root programs now plan a separate post-quantum trust path built on Merkle trees rather than on ordinary certificate chains [4] [3].

The third is a deadline that bit. The first phase of the SHA-1 sunset took effect on 15 September [5]. Within three days, four certificate authorities had incident reports open for failing to revoke SHA-1 signed subordinate certificates on time [6].

The fourth is a ballot caught between two states. SC104 turns the Authority Information Access extension in subscriber certificates from a requirement into a recommendation [7]. The working group chair announced on 4 October that it became effective that day, attaching Baseline Requirements version 2.3.1 [8]. At the last check that day, the Forum's requirements page still showed 2.3.0, and no 2.3.1 release existed in its repository [9] [10].

2. A New Kind of Certificate Authority

A Merkle Tree Certificate replaces the usual chain of signatures with proof that the certificate appears in a signed public log, which keeps large post-quantum signatures out of most connections.

Recent Developments

Three root programs and one prospective issuer set out their post-quantum plans in the weeks around this update. Chrome's came first. Its draft quantum-resistant root program policy, version 0.3.0 from 14 August, describes a root store made up entirely of Merkle Tree Certificate authorities and independent mirroring operators [3].

The draft is specific about hardware and time. Cosigner keys must be ML-DSA-44, held in modules validated to FIPS 140-3 Level 3 or Common Criteria EAL 4+ [3]. Subscriber certificates may last at most seven days [3].

One clause shapes who can issue first. The draft limits initial eligibility to organisations that operated a usable Certificate Transparency log before 1 February 2026 [3]. Chrome opened applications for its testing programme on 31 July, with a production phase planned for 2027 [2].

Apple's plan has the same shape with a harder edge. Each certificate would need three cosignatures [4]. The issuer provides one with ML-DSA and one with ECDSA, and a different operator acting as a mirror adds a third with ML-DSA [4]. Keys must sit in the same grade of hardware, validity is capped at seven days, and operators face annual audits [4]. Applications to run such an authority open in late summer 2027 [4].

Apple also opened a second door. It will accept composite roots that pair ML-DSA with ECDSA or RSA for S/MIME email [4].

Cloudflare's announcement shows what an issuer of this kind looks like in practice. During its experiment with Chrome it ran a bootstrap authority whose certificates went to half of Chrome Beta 146 users [1]. When a client could not negotiate the new format, the connection fell back to the traditional chain [1]. The company says the experiment wound down in August after serving billions of certificates [1].

Its production authority will run a fork of Boulder, the ACME software Let's Encrypt uses [1]. It will also operate mirroring cosigners and launch a new family of static transparency logs [1]. Under these designs, an issuer has to run logs and mirrors as well as sign certificates. Chrome's testing programme already requires every participant to operate both [2].

Version 06 of the IETF Merkle Tree Certificates draft was published on 21 September, by authors from Google, Apple and Cloudflare among others [11].

Microsoft took a different route. On 27 August it deployed seven ML-DSA-87 pilot roots for approved authorities, including DigiCert, Sectigo, SSL.com and a Greek academic authority [12]. These roots are for controlled testing only and are not publicly trusted [12]. They will not be added to the CCADB or to transparency logs [13]. Microsoft's pilot keeps the conventional chain and changes only the algorithm.

Current State

None of these plans is yet a published rule. Apple's root program repository had no commit after 4 August, so its draft policy was not out as of 4 October [14]. Chrome's draft remained at version 0.3.0 [15].

The debate has already started. On 29 September a participant on the transparency policy list objected to the seven-day cap [4]. In that view, a maximum lifetime that short is too big a change to bring in as part of Merkle certificates, and should be announced separately [4].

3. Post-Quantum Ballots Inside the Forum

The root programs can build a new trust path on their own, but ordinary certificates still need the Forum's rules to allow post-quantum keys.

Current State

The Forum's two post-quantum ballots stand at very different points. The web ballot now has a number. SC-106 would permit ML-DSA keys and signatures in all three parameter sets, in server certificates, issuing authorities, revocation lists and status responses [16]. Amazon Trust Services took over its leadership from Google Trust Services in late August [17].

The draft requires pure post-quantum chains, so an ML-DSA key must be certified by an ML-DSA signature [16]. Mozilla and Chrome representatives objected to that restriction [17]. They also argued that the proposal reaches beyond the group's browser focus, and suggested a separate working group for other certificate consumers [16]. The 10 September call recorded no updates, and the pull request has not changed since 31 August [18] [16].

The email ballot is moving. SMC019, opened on 24 September, would add composite ML-DSA to the S/MIME requirements [19]. A composite pairs a post-quantum signature with a traditional one. It builds on the standalone ML-DSA and ML-KEM support already published through SMC013 [19].

Neither ballot has formally started. No discussion period has been announced for SMC019, and it does not yet appear on its group's ballots page [20]. The email ballot also waits on the IETF. The composite signatures draft for X.509 sits in the RFC Editor queue and is not yet an RFC [21].

SMC019 cites Internet-Drafts for now, with a note to update once RFC numbers are assigned [19].

4. What Took Effect in September

A ballot binds no one until it is published and its effective date arrives, and September was a month when several dates arrived together.

Recent Developments

A cluster of deadlines landed in the middle of the month. The first came on 7 September, when Baseline Requirements version 2.3.0 took effect [9]. It carries Ballot SC100, which clarifies and consolidates the rules on DNSSEC [9].

The SHA-1 sunset followed on 15 September. From that date, certificate authorities may not sign new revocation lists with SHA-1, and must revoke any unexpired subordinate certificate carrying a SHA-1 signature [5]. The ballot had passed with 26 issuer votes and all four major root programs in favour [5].

Four authorities missed it. GoDaddy revoked two subordinate certificates from 2006 two days late, and DigiCert revoked two issuing authorities after a third party reported them [6] [22]. Firmaprofesional and FNMT each had a SHA-1 subordinate left unrevoked past the deadline [22].

The same day brought three more dates. Code Signing Baseline Requirements version 3.11.0 took effect, making a reserved policy identifier mandatory in subscriber certificates [23]. New S/MIME root and subordinate RSA keys must now be at least 4096 bits [24]. The CCADB policy began requiring every authority's policy documents to state adherence to it [25]. Enforcement bugs followed within days [26].

Email got a new version as well. Ballot SMC018 was published as S/MIME Baseline Requirements version 1.0.16 on 28 September, four days after its review ended [27]. From 1 July 2027, S/MIME subordinate and subscriber certificates may carry only the email protection purpose, plus client authentication as an option [24].

Current State

SC104 is the one item still between stages. It passed on 3 September with 21 issuer votes and five certificate consumer votes, and none against [7]. Its redline was merged into the repository on the morning of 4 October [28]. The chair says it is effective [8]. The Forum's ballots page still showed it in review that day, and the requirements page still ended at 2.3.0 [29] [9].

5. Rules Still Being Written

Most of the Forum's work at any moment is proposals under discussion, and these show where next year's requirements will come from.

Recent Developments

The most consequential live proposal is about what revocation means. SC-107, proposed by Sectigo and endorsed by Google and DigiCert, opened for discussion on 17 September [30]. A problem report would need at least a serial number or hash, plus a described violation or a revocation reason [30]. The authority would have 24 hours to decide whether the report is actionable [30].

The ballot also defines the word itself. A certificate counts as revoked when its serial number appears in a revocation list or draws a revoked status response, from 15 February 2027 [30]. Discussion was due to end on 1 October. A post appeared in the thread that day, and no voting thread existed by 4 October [31].

Other server ballots are slower. SC103 made no progress, and SC105, removing “High Risk” language, is still a draft [18]. A draft omnibus would move the extended validation profile into the Baseline Requirements and turn the EV Guidelines into a validation methods document [32].

The other groups have drafts of their own. SMC020 would drop the rule to record which requirements version a validation method came from [33]. CSC-034 would clarify name fields for individual code signing subjects [34]. CSC-033 has sat in discussion since May with no vote called [35].

Validation work is looking ahead. A revised ACME persistent DNS method defends against malicious proxy servers [36]. The Validation Subcommittee is moving toward informative guidance on artificial intelligence, perhaps as an appendix, modelled on an accounting standards document [37].

Governance moved as well. A special election filled the Forum chair and five working group chairs, and a tied network security vote was settled by agreement between the two candidates [38].

The public list also discussed a charter for a client authentication working group, with emerging consensus to proceed [39]. Its scope is mutual TLS for people, not email, and no ballot exists yet [39]. The timing fits Chrome’s policy. New subordinates in its store had to be server-only from June 2026, and new end-entity certificates follow in March 2027 [40].

Unresolved Conflicts

One disagreement sits in this section. NS-010, which restructures the network security requirements, held its vote from 10 to 17 August [41]. No result has been announced, and the Forum’s site still lists it as voting [42].

An earlier account described an August network security ballot adding a section on infrastructure inventory, which does not match NS-010’s stated content. A posted result with its final text would show whether these are one ballot or two.

6. Who Gets Trusted

Root programs decide which authorities browsers trust, so their admissions, their policies and the incidents they police show how the existing issuers are faring.

Recent Developments

New roots are queuing for trust while Cloudflare prepares to apply. GoDaddy's two new web roots cleared public discussion on 29 September with no objections [43]. Mozilla posted its intent to approve on 2 October, with last call to 9 October [44]. It then deferred any approval until at least seven days after GoDaddy files an updated practice statement [44].

eMudhra's four email and web roots reached the same stage [45]. One participant questioned eMudhra's recent incident performance, and asked that such notices list a candidate's incidents [45]. A separate thread asked Mozilla to say what action it considers appropriate over NETLOCK's incident handling [46]. An inclusion request for two new ISRG roots, from the organisation behind Let's Encrypt, opened on 22 September [47].

The incident load was heavy. Thirty-four compliance bugs opened between 12 and 26 September [22]. Nineteen more followed by 2 October, twelve of them about revocation lists, including bugs against Apple and GoDaddy [48]. Others covered undisclosed issuing authorities at DigiCert and certificates issued to sanctioned entities at Sectigo [48]. No program announced a distrust or removal [49].

Transparency gained a new operator. GoDaddy's five new tiled logs became qualified in Chrome on 30 September [50].

The policies kept tightening. Mozilla published a whitepaper on the Detailed Controls Reports its policy will require [51]. It also merged a template for mass revocation plans, with thresholds of 100 certificates or 1% of active web certificates [52].

Unresolved Conflicts

Two disagreements sit in this section. On mass revocation, one source says plans must be tested each year by revoking 30 random certificates within five days, with attestation from April 2026 [53]. Mozilla's template and its earlier guidance date third-party assessment from the June 2025 audit cycle [52] [54]. The current text of the policy's revocation section would settle it.

The WebTrust applicability matrix lists Baseline Requirements versions 2.2.5 and 2.10 as current [55]. Neither matches the published sequence. A refreshed matrix would settle it.

7. Where Things Stood Before This Update

Readers who skipped the earlier editions need the starting point, and this section gives it.

Before September, the picture was one of steady tightening with post-quantum work still at the margins. The current web requirements were version 2.2.9, effective 6 August [56]. SC104 was in its vote, SC103 in discussion and SC105 still a draft [57].

The largest change in force was the shrinking certificate lifetime. Certificates issued from 15 March 2026 could last at most 200 days, falling to 100 days in March 2027 and 47 days in March 2029 [58]. Domain validation from several network vantage points was arriving in stages, with five perspectives due by December 2026 [59].

Post-quantum keys were not allowed in web certificates. Draft text for ML-DSA existed only as a pull request [60]. The working group had named two blockers, its own requirements and the IETF's encoding standards [61]. The email group was ahead, having published post-quantum algorithms through SMC013 in August 2025 [62].

The root programs were already demanding more than the Forum. Mozilla's policy 3.1 took effect in July and will require Detailed Controls Reports from audit periods starting July 2027 [63]. Chrome's policy 1.8 limited roots per owner and pushed dedicated hierarchies [64]. Microsoft's new requirements took effect in July and Apple's in August [65] [66].

Trust removal came in two kinds. DigiCert's oldest roots were retired in April 2026 for age [67]. The punitive precedent remained the 2024 distrust of Entrust [68].

Revocation was being routed around rather than repaired. Let's Encrypt ended its status responder service in August 2025 [69]. Firefox's CRLite covered 99.84% of logged certificates [70].

Transparency was changing shape too. Chrome had begun accepting the static log design [71]. The Merkle Tree Certificates draft stood at version 05 in July [72]. In September, that draft became the centre of the story.

Sources

1. Building a post-quantum certificate authority with Merkle Tree Certificates - Cloudflare Blog - <https://blog.cloudflare.com/pq-ca-with-mtcs/> - as of 2026-09-29
2. Chrome Quantum-resistant Root Program, MTC testing - Google Chrome - <https://googlechrome.github.io/chromerootprogram/cqrp/testing> - as of 2026-09-26
3. [DRAFT] Chrome Quantum-resistant Root Program Policy, Version 0.3.0 - Google Chrome - <https://googlechrome.github.io/chromerootprogram/cqrp/draft-policy> - as of 2026-08-14
4. Apple Root Program PQC Announcement - chromium ct-policy list - <https://groups.google.com/a/chromium.org/g/ct-policy/c/QGZw2ADMXvk> - as of 2026-09-29
5. Ballot SC097, Sunset all remaining use of SHA-1 signatures in certificates and CRLs - CA/Browser Forum - <https://cabforum.org/2026/01/24/ballot-sc097-sunset-all-remaining-use-of-sha-1-signatures-in-certificates-and-crls/> - as of 2026-09-15
6. Bug 2072663, GoDaddy delayed revocation of SHA-1 signed subordinate CA certificates - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2072663 - as of 2026-09-18
7. Ballot SC104, Set presence of AIA extension to SHOULD for Subscriber Certificates - CA/Browser Forum - <https://cabforum.org/2026/09/03/ballot-sc104-set-presence-of-aia-extension-to-should-for-subscriber-certificates/> - as of 2026-09-03
8. Ballot SC104 IPR review results and effective date - servercert-wg list - <https://groups.google.com/a/groups.cabforum.org/g/servercert-wg/c/3cQWf7MiTBg> - as of 2026-10-04
9. Latest Baseline Requirements - CA/Browser Forum - <https://cabforum.org/working-groups/server/baseline-requirements/requirements/> - as of 2026-10-04

10. cabforum/servercert releases - GitHub - <https://api.github.com/repos/cabforum/servercert/releases> - as of 2026-10-04
11. draft-ietf-plants-merkle-tree-certs-06 - IETF Datatracker - <https://datatracker.ietf.org/doc/draft-ietf-plants-merkle-tree-certs/> - as of 2026-09-21
12. Microsoft Trusted Root Program announcements, PQC TLS pilot roots - TrustedRootProgram - <https://github.com/TrustedRootProgram/Program-Requirements/blob/main/Announcements.md> - as of 2026-08-27
13. Post-Quantum Cryptography (PQC) TLS Pilot Program Requirements V1.0 - TrustedRootProgram - <https://github.com/TrustedRootProgram/Program-Requirements/blob/main/PQC%20Pilot%20Program.md> - as of 2026-09-26
14. apple-root-program repository, commit history - Apple - <https://github.com/apple/apple-root-program/commits/main> - as of 2026-10-04
15. chromerootprogram repository, commit history - Google Chrome - <https://github.com/GoogleChrome/chromerootprogram/commits/main> - as of 2026-09-24
16. Pull request 679, SC-106 Enable Post-Quantum Cryptography (PQ) Key Pairs in TLS Server Certificates - cabforum/servercert - <https://github.com/cabforum/servercert/pull/679> - as of 2026-08-31
17. Minutes of the Server Certificate Working Group, 27 August 2026 - CA/Browser Forum - <https://cabforum.org/2026/08/27/2026-08-27-minutes-of-the-server-certificate-working-group/> - as of 2026-08-27
18. Minutes of the Server Certificate Working Group, 10 September 2026 - CA/Browser Forum - <https://cabforum.org/2026/09/10/2026-09-10-minutes-of-the-server-certificate-working-group/> - as of 2026-09-10
19. Pull request 310, SMC019 Introduction of Composite ML-DSA Algorithms - cabforum/smime - <https://github.com/cabforum/smime/pull/310> - as of 2026-09-24
20. S/MIME Certificate Working Group Ballots - CA/Browser Forum - <https://cabforum.org/working-groups/smime/ballots/> - as of 2026-10-04
21. draft-ietf-lamps-pq-composite-sigs - IETF Datatracker - <https://datatracker.ietf.org/doc/draft-ietf-lamps-pq-composite-sigs/> - as of 2026-10-04
22. CA Certificate Compliance bugs opened from 12 September 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Compliance&creation_time=2026-09-12 - as of 2026-09-26
23. Code Signing Baseline Requirements, current version and history - CA/Browser Forum - <https://cabforum.org/working-groups/code-signing/requirements/> - as of 2026-09-26
24. Latest S/MIME Baseline Requirements - CA/Browser Forum - <https://cabforum.org/working-groups/smime/requirements/> - as of 2026-09-28
25. CCADB Policy, Version 2.1 - CCADB - <https://www.ccadb.org/policy> - as of 2026-09-15
26. Bug 2073388, SHECA CP/CPS missing root program adherence statements - Mozilla Bugzilla - https://bugzilla.mozilla.org/show_bug.cgi?id=2073388 - as of 2026-09-23

27. Ballot SMC018 adoption and S/MIME BR 1.0.16 publication notice - smcwg-public list - https://groups.google.com/a/groups.cabforum.org/g/smcwg-public/c/iZ_yg8O0-mA - as of 2026-09-28
28. Pull request 665, SC-104 redline - cabforum/servercert - <https://api.github.com/repos/cabforum/servercert/pulls/665> - as of 2026-10-04
29. Server Certificate Working Group Ballots - CA/Browser Forum - <https://cabforum.org/working-groups/server/ballots/> - as of 2026-10-04
30. Discussion Period Starts: SC-107 Improve Certificate Problem Reports and Clarify the Meaning of Revocation - servercert-wg list - https://groups.google.com/a/groups.cabforum.org/g/servercert-wg/c/ZzkzUhjxI_0 - as of 2026-09-26
31. servercert-wg list, thread index - CA/Browser Forum - <https://groups.google.com/a/groups.cabforum.org/g/servercert-wg> - as of 2026-10-04
32. Pull request 680, SC-1XX EVG Cleanup and BR Sync - cabforum/servercert - <https://github.com/cabforum/servercert/pull/680> - as of 2026-09-29
33. Pull request 309, SMC020 Remove requirement to record BR version of validation method - cabforum/smime - <https://github.com/cabforum/smime/pull/309> - as of 2026-09-23
34. Pull request 59, CSC-034 Clarify Certificate Profile for Individuals - cabforum/code-signing - <https://github.com/cabforum/code-signing/pull/59> - as of 2026-09-24
35. Ballot CSC-033, 2026-05 SCWG alignment ballot - cscwg-public list - https://groups.google.com/a/groups.cabforum.org/g/cscwg-public/c/erLQwy_gM8w - as of 2026-05-26
36. New version of ACME dns-persist-01 with record changes - Validation Subcommittee list - <https://groups.google.com/a/groups.cabforum.org/g/validation/c/9fucAuWGQdo> - as of 2026-09-26
37. Approach for AI in validation - Validation Subcommittee list - <https://groups.google.com/a/groups.cabforum.org/g/validation/c/56NM0wHv9vo> - as of 2026-10-01
38. Special election results - CA/Browser Forum public list - https://groups.google.com/a/groups.cabforum.org/g/public/c/M_iG_HzoPXs - as of 2026-09-24
39. clientAuth Working Group charter discussion - CA/Browser Forum public list - <https://groups.google.com/a/groups.cabforum.org/g/public/c/m6yFd9cyUes> - as of 2026-09-24
40. Chrome Root Program Policy v1.8: Full 2026-2027 Timeline - Encryption Consulting - <https://www.encryptionconsulting.com/chromes-root-program-policy-v1-8/> - as of 2026-09-12
41. NS-010 v2 voting thread - netsec list - <https://groups.google.com/a/groups.cabforum.org/g/netsec/c/W9zx0BvXxw0> - as of 2026-09-09
42. Network Security Working Group Ballots - CA/Browser Forum - <https://cabforum.org/working-groups/netsec/ballots/> - as of 2026-09-27

43. Public discussion of GoDaddy root inclusion request - CCADB public list - <https://groups.google.com/a/ccadb.org/g/public/c/6hPxKjObVlc> - as of 2026-09-29
44. Notice of Intent to Approve, GoDaddy root inclusion - mozilla dev-security-policy - <https://groups.google.com/a/mozilla.org/g/dev-security-policy/c/XIUPVy2F0gs> - as of 2026-10-02
45. Notice of Intent to Approve, eMudhra root inclusion - mozilla dev-security-policy - <https://groups.google.com/a/mozilla.org/g/dev-security-policy/c/h2ucXHS5HZ4> - as of 2026-10-02
46. Netlock inclusion in MRSP - mozilla dev-security-policy - <https://groups.google.com/a/mozilla.org/g/dev-security-policy/c/a7d2tpkX-A> - as of 2026-10-02
47. CA Certificate Root Program bugs changed from 25 September 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Root%20Program&last_change_time=2026-09-25&include_fields=id,summary,creation_time,last_change_time,status,resolution - as of 2026-10-04
48. CA Certificate Compliance bugs opened from 27 September 2026 - Mozilla Bugzilla - https://bugzilla.mozilla.org/rest/bug?product=CA%20Program&component=CA%20Certificate%20Compliance&creation_time=2026-09-27&include_fields=id,summary,creation_time,status - as of 2026-10-02
49. mozilla dev-security-policy list - Mozilla - <https://groups.google.com/a/mozilla.org/g/dev-security-policy> - as of 2026-10-04
50. Chrome Certificate Transparency log list, v3 - Google - https://www.gstatic.com/ct/log_list/v3/log_list.json - as of 2026-10-03
51. Pull request 303, Detailed Controls Reports whitepaper Version 1.0 - mozilla/pkipolicy - <https://github.com/mozilla/pkipolicy/pull/303> - as of 2026-09-23
52. Mass Revocation Template - mozilla/pkipolicy - <https://raw.githubusercontent.com/mozilla/pkipolicy/master/rootstore/Mass-Revocation-Template.md> - as of 2026-09-23
53. Add section 6.1.3 for Delayed Revocation - mozilla/pkipolicy - <https://github.com/mozilla/pkipolicy/commit/efa8ac40ac341fb813620938ef72328a53858038> - as of 2026-09-12
54. Responding to a misissuance, mass revocation requirements - MozillaWiki - https://wiki.mozilla.org/CA/Responding_To_A_Misissuance - as of 2025-09-01
55. Principles and criteria and practitioner guidance - CPA Canada - <https://www.cpacanada.ca/business-and-accounting-resources/audit-and-assurance/overview-of-WebTrust-services/principles-and-criteria> - as of 2026-08-01
56. Latest Baseline Requirements, revision table - CA/Browser Forum - <https://cabforum.org/working-groups/server/baseline-requirements/requirements/> - as of 2026-08-06
57. Server Certificate Working Group Ballots index - CA/Browser Forum - <https://cabforum.org/working-groups/server/ballots/> - as of 2026-09-01

58. Ballot SC081v3, Introduce Schedule of Reducing Validity and Data Reuse Periods - CA/Browser Forum - <https://cabforum.org/2025/04/11/ballot-sc081v3-introduce-schedule-of-reducing-validity-and-data-reuse-periods/> - as of 2026-09-01
59. MPIC FAQ - Sectigo - <https://www.sectigo.com/mpic-faq> - as of 2026-09-01
60. Pull request 662, permit ML-DSA public keys and signatures - cabforum/servercert - <https://github.com/cabforum/servercert/pull/662> - as of 2026-09-01
61. Minutes of the Server Certificate Working Group, 21 May 2026 - CA/Browser Forum - <https://cabforum.org/2026/05/21/2026-05-21-minutes-of-the-server-certificate-working-group/> - as of 2026-05-21
62. Ballot SMC013, Introduction of PQC Algorithms - CA/Browser Forum - <https://cabforum.org/working-groups/smime/requirements/> - as of 2025-08-22
63. Improving Transparency and Assurance in the Web PKI, Mozilla Root Store Policy v3.1 - Mozilla Security Blog - <https://blog.mozilla.org/security/2026/06/29/improving-transparency-and-assurance-in-the-web-pki-mozilla-root-store-policy-v3-1/> - as of 2026-07-01
64. Chrome Root Program Policy, Version 1.8 - Google Chrome - <https://googlechrome.github.io/chromerootprogram/crp/policy> - as of 2026-08-28
65. Microsoft Root Program Requirements - TrustedRootProgram - <https://github.com/TrustedRootProgram/Program-Requirements/blob/main/Requirements.md> - as of 2026-09-26
66. Apple Root Program Policy - Apple - <https://raw.githubusercontent.com/apple/apple-root-program/main/policy.md> - as of 2026-09-12
67. DigiCert G1 root removal advisory - DigiCert - <https://docs.digicert.com/en/certcentral/order-and-manage-certificates/reports-and-advisories/review-the-g1-root-removal-advisory.html> - as of 2026-04-15
68. Sustaining Digital Certificate Security, Entrust Certificate Distrust - Google Online Security Blog - <https://security.googleblog.com/2024/06/sustaining-digital-certificate-security.html> - as of 2024-11-12
69. OCSP Service Has Reached End of Life - Let's Encrypt - <https://letsencrypt.org/2025/08/06/ocsp-service-has-reached-end-of-life> - as of 2025-08-06
70. CRLite in production in Firefox - Mozilla Hacks - <https://hacks.mozilla.org/2025/08/crlite-fast-private-and-comprehensive-certificate-revocation-checking-in-firefox/> - as of 2025-08-19
71. Chrome CT Policy changes to accept static-ct-api logs - chromium ct-policy list - <https://groups.google.com/a/chromium.org/g/ct-policy/c/HBFZHG0TCsY/m/HAAVRK6MAAAJ> - as of 2026-04-01
72. draft-ietf-plants-merkle-tree-certs-05 - IETF Datatracker - <https://datatracker.ietf.org/doc/draft-ietf-plants-merkle-tree-certs/> - as of 2026-07-06